Hypocoercivity-preserving discontinuous Galerkin methods for kinetic equations

Application to the Vlasov-Poisson-Fokker-Plack system

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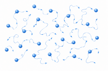
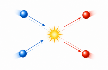
Kinetic theory: bridging microscopic and macroscopic world

Microscopic

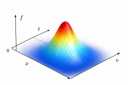


Motion



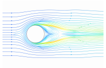
Interaction

Kinetic

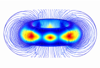


Distribution

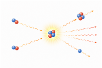
Macroscopic



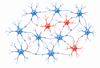
Rarefied gas



Plasma



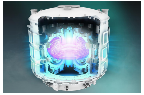
Radiation



Collective dynamics

Applications

Aerospace



Controlled fusion

Active matter

Linear kinetic equation conserving mass

General model

$$\partial_t f + \mathscr{T}(f) = \mathscr{C}(f).$$

- distribution function $f(t,x,v)$.
- time $t \geq 0$, dimension $d \geq 1$.
- position $x \in \mathcal{X} \subset \mathbb{R}^d$.
- velocity $v \in \mathcal{V} \in \{\mathbb{S}^{d-1}, \mathbb{R}^d\}$.
- orthogonal projection Π onto $\ker(\mathscr{C})$.
- potential $\Phi(x)$.
- equilibrium $\mathcal{M}(v)$ with $\int \mathcal{M} \mathrm{d}v = 1$.

Transport

$$\mathscr{T}(f) = v \cdot \nabla_x f - \nabla_x \Phi(x) \cdot \nabla_v f.$$

Collision

(H1) $\mathscr{C}$ is **linear** and acts only on v.

(H2) $\mathscr{C}$ is **self-adjoint** and **microscopic coercive** in $L^2_v(\mathcal{M}^{-1})$, namely, $\mathscr{C} = \mathscr{C}^*$ and

$$-\langle \mathscr{C}(f), f \rangle_{L^2_v(\mathcal{M}^{-1})} \geq \lambda_{\mathscr{C}} \| f - \Pi(f) \|^2_{L^2_v(\mathcal{M}^{-1})}.$$

(H3) $\mathscr{C}$ has a **one-dimensional** kernel of **collision invariants** in $L^2_v(\mathcal{M}^{-1})$, namely,

$$\ker(\mathscr{C}) = \operatorname{span}\{\mathcal{M}(v)\} \quad \text{and} \quad \int \mathscr{C}(f) \mathrm{d}v = 0.$$

Examples

Typical collisions

Let $\mathcal{V} = \mathbb{R}^d$ and $\mathcal{M}(v) = (2\pi)^{-\frac{d}{2}} e^{-\frac{1}{2}|v|^2}$,

- **Fokker-Planck** type:

$$\mathscr{C}(f) = \nabla_v \cdot (vf + \nabla_v f).$$

- **BGK** type: mass density $\rho = \int f \mathrm{d}v$,

$$\mathscr{C}(f) = \rho\mathcal{M} - f.$$

Let $\mathcal{V} = \mathbb{S}^{d-1}$ and $\mathcal{M}(v) \equiv |\mathcal{V}|^{-1}$,

- **Isotropic scattering** type:

$$\mathscr{C}(f) = \rho\mathcal{M} - f.$$

Gibbs state

Characterize

$$f_\infty(x, v) \in \ker(\mathscr{T}) \cap \ker(\mathscr{C}).$$

by initial data and conservation of total mass

$$f_\infty(x, v) = c_\infty e^{-\Phi(x)} \mathcal{M}(v),$$

where c_∞ is given by

$$c_\infty \int e^{-\Phi(x)} \mathrm{d}x = \int \rho^{in}(x) \mathrm{d}x.$$

Quantitative long-time behavior

Long-time limit

$$f(t,x,v) \xrightarrow{t\to+\infty} f_\infty(x,v).$$

Compactness methods[1]: not quantitative

Missing ingredient

Collision only dissipates microscopic variables, **transport propagates the microscopic dissipation to macroscopic variables**.

Challenge

Standard energy estimate

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|f(t)-f_\infty\|^2_{L^2_{x,v}(f_\infty^{-1})} = \langle \mathscr{C}(f), f\rangle_{L^2_{x,v}(f_\infty^{-1})} \leq 0.$$

Microscopic coercivity implies

$$\langle \mathscr{C}(f), f\rangle_{L^2_{x,v}(f_\infty^{-1})} = 0 \iff f = \rho\mathcal{M}.$$

Lack of control between ρ and ρ_∞!

Hypocoercivity method

There exist $C > 0$ and $\lambda > 0$ such that[2]

$$\|f(t)-f_\infty\|_{L^2_{x,v}(f_\infty^{-1})} \leq Ce^{-\lambda t}\|f^{in}-f_\infty\|_{L^2_{x,v}(f_\infty^{-1})}.$$

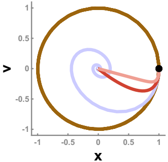
[1] Bouchut and Dolbeault (1995)

[2] Villani (2009); Dolbeault, Mouhot, and Schmeiser (2015)

Hypocoercivity: damping oscillator

Model description

$$\begin{cases} \dfrac{dx}{dt} = +v, \\ \dfrac{dv}{dt} = -x - \beta v. \end{cases}$$



Energy $H = \frac{1}{2}(|x|^2 + |v|^2)$ satisfies

$$\frac{dH}{dt} = -\beta|v|^2 \le 0.$$

Observation: lack of dissipation in x!

Key idea: modify energy by adding a **correction term** to recover missing dissipation

$$G = H + \alpha xv.$$

Hypocoercive estimate

$$\frac{dG}{dt} = -\beta|v|^2 + \alpha|v|^2 - \alpha|x|^2 - \alpha\beta xv$$
$$\le -\left(\beta - \alpha - \frac{\beta^2}{2}\alpha\right)|v|^2 - \frac{\alpha}{2}|x|^2.$$

Taking $\alpha = \frac{2\beta}{3+\beta^2}$ and $\lambda = \frac{\alpha}{2(1+\alpha)}$, it yields

- **Equivalence**: $(1-\alpha)H \le G \le (1+\alpha)H$;
- **Exponential decay**: $\dfrac{dG}{dt} \le -\lambda G$.

Let $C = \frac{1+\alpha}{1-\alpha}$, it follows that

$$H(t) \le Ce^{-\lambda t}H(0).$$

Hypocoercivity: linear kinetic equation

Model description

$$\partial_t f + v\partial_x f = \rho\mathcal{M} - f.$$

Velocity integration after multiplying 1 and v,

$$\begin{cases} \partial_t \rho + \partial_x j = 0, \\ \partial_t j + D\partial_x rho + R = -j, \end{cases}$$

where $j = \int v f \,\mathrm{d}v$, $D = \frac{1}{d}\int |v|^2 \mathcal{M} \mathrm{d}v$ and

$$R = \int v^2\partial_x)(f - \rho\mathcal{M})\mathrm{d}v.$$

Standard energy estimate

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|f(t) - f_\infty\|^2_{L^2_{x,v}(f_\infty^{-1})} = -\|f - \rho\mathcal{M}\|^2_{L^2_{x,v}(f_\infty^{-1})}.$$

Key idea

Modified energy[3]:

$$\mathcal{H} = \frac{1}{2}\|f - f_\infty\|^2_{L^2_{x,v}(f_\infty^{-1})} + \alpha\langle\nabla_x \cdot j, \psi\rangle_{L^2_x(\rho_\infty^{-1})},$$

where

$$D\Delta_x\psi = \rho - \rho_\infty.$$

[3] Bessemoulin-Chatard, Herda, and Rey (2020)

Hypocoercivity: linear kinetic equation

Regularity estimates[4]

$$\|j\|_{L^2_x(\rho_\infty^{-1})} + \|R\|_{L^2_x(\rho_\infty^{-1})} \lesssim \|f - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})},$$
$$\|\psi\|_{L^2_x(\rho_\infty^{-1})} + \|\partial_x\psi\|_{L^2_x(\rho_\infty^{-1})} \lesssim \|\rho - \rho_\infty\|_{L^2_x(\rho_\infty^{-1})}.$$

Computation

$$\frac{d\mathcal{H}}{dt} = -\|f - \rho\mathcal{M}\|^2_{L^2_{x,v}(f_\infty^{-1})} + \alpha(\mathcal{J}_1 + \mathcal{J}_2 + \mathcal{J}_3)$$
$$\mathcal{J}_1 = -D\langle\Delta_x\rho, \psi\rangle_{L^2_x(\rho_\infty^{-1})} = -\|\rho - \rho_\infty\|^2_{L^2_x(\rho_\infty^{-1})}$$
$$\mathcal{J}_2 = -\langle\nabla_x\cdot j, \psi\rangle_{L^2_x(\rho_\infty^{-1})} \lesssim +\|f - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})}\|\rho - \rho_\infty\|_{L^{-1}_x(\rho_\infty^{-1})}$$
$$\mathcal{J}_3 = -\langle\nabla_x\cdot R, \psi\rangle_{L^2_x(\rho_\infty^{-1})} \lesssim +\|f - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})}\|\rho - \rho_\infty\|_{L^{-1}_x(\rho_\infty^{-1})}$$

Hypocoercive estimate

By Young's inequality, there exists $C > 0$ such that for all $\alpha \in [0, 1/2C]$

$$\frac{1}{2}\mathcal{E} \leq \mathcal{H} \leq \frac{3}{2}\mathcal{E},$$

and

$$\frac{d\mathcal{H}}{dt} \leq -\frac{\alpha}{2}\|\rho - \rho_\infty\|^2_{L^2_x(\rho_\infty^{-1})} - (1 - 2\alpha C)\|f - \rho\mathcal{M}\|^2_{L^2_{x,v}(f_\infty^{-1})}.$$

Take $\alpha = 2/(4C + 1)$, then

$$\frac{d\mathcal{H}}{dt} \leq -\frac{2}{3}\alpha\mathcal{H}.$$

[4] Dolbeault, Mouhot, and Schmeiser (2015); Bessemoulin-Chatard, Herda, and Rey (2020)

Hypocoercivity: diffusive limit

Model description

Kinetic model:

$$\varepsilon\partial_t f + v\cdot\nabla_x f = \frac{1}{\varepsilon}(\rho\mathcal{M} - f).$$

Diffusion model:

$$\partial_t \rho^0 = D\Delta_x \rho^0.$$

$$\begin{tikzcd}[column sep=large]
f(t,x,v) \arrow[rr, "t/\varepsilon \,\to\, +\infty"] \arrow[dr, "\varepsilon \,\to\, 0"'] & & \rho_\infty(x)\mathcal{M}(v) \\
& \rho^0(t,x)\mathcal{M}(v) \arrow[ur, "t \,\to\, +\infty"'] &
\end{tikzcd}$$

Hypocoercive estimate

There are constants $C_1, C_2 \geq 1$ and $\lambda_1, \lambda_2 > 0$ such that for all $\varepsilon \in (0,1)$, the solution satisfies[5]:

$$\|f(t) - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})} \leq e^{-\lambda_1 t/(4\varepsilon^2)}\|f(0) - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})} + \varepsilon C_1 e^{-\lambda_1 t}\|\rho(0) - \rho_\infty\|_{H^1_x(\rho_\infty^{-1})},$$

and

$$\|\rho(t) - \rho^0(t)\|_{H^{-1}_x(\rho_\infty^{-1})} \leq C_2 e^{-\lambda_2 t}\left(\|\rho(0) - \rho^0(0)\|_{H^{-1}_x(\rho_\infty^{-1})} + \varepsilon\|\rho(0) - \rho_\infty\|_{H^1_x(\rho_\infty^{-1})}\right)$$

[5] Blaustein and Filbet (2024)

Contents

Nonlinear Vlasov-Poisson-Fokker-Planck model

1D VPFP model

$$\partial_t f + v\partial_x f + E\partial_v f = \frac{1}{\tau_0}\partial_v\left(vf + T_0\partial_v f\right),$$

$$E = -\partial_x \Phi, \quad -\partial_{xx}\Phi = \rho - 1.$$

- electric potential $\Phi(t, x)$.
- electric field $E(t, x)$.
- background temperature $T_0 > 0$.
- mean free time $\tau_0 > 0$.

Equilibria

Maxwellian:

$$\mathcal{M}(v) = (2\pi T_0)^{-\frac{1}{2}} e^{-\frac{|v|^2}{2T_0}}.$$

Stationary state:

$$f_\infty(v) = \mathcal{M}(v), \quad \rho_\infty \equiv 1, \quad E_\infty \equiv 0.$$

Long-time behavior

Continuous framework

- Compactness methods[6]: not quantitative
- Perturbative methods[7]: quantitative with close-to-equilibrium assumption

Discrete framework

First-order finite difference/volume method:

- Linear VFP equation[8]: analysis and algorithm
- Nonlinear VPFP system[9]: analysis for **linearized** model and algorithm for nonlinear model

Our goals

- **Discontinuous Galerkin** method.
- analysis for **nonlinear** VPFP system.

[6] Soler, Carrillo, and Bonilla (1997)

[7] Herda and Rodrigues (2018); Addala et al. (2021)

[8] Blaustein and Filbet (2024), Math. Comput.

[9] Blaustein and Filbet (2024), J. Comput. Phys.

Close-to-equilibrium framework: linearized free energy

Logarithmic free energy estimate

$$\frac{d\mathcal{G}}{dt} = -\frac{1}{\tau_0}\mathcal{D}.$$

- logarithmic free energy

$$\mathcal{G} = \int f \log\left(\frac{f}{f_\infty}\right) dxdv + \frac{1}{2T_0}\|E\|_{L_x^2}^2.$$

- dissipation from collision

$$\mathcal{D} = 4T_0 \int \left|\partial_v\left(\sqrt{\frac{f}{f_\infty}}\right)\right|^2 f_\infty dxdv.$$

- **Key idea**: linearization near f_∞

$$f \log\left(\frac{f}{f_\infty}\right) \approx f - f_\infty + \frac{|f - f_\infty|^2}{2f_\infty}$$

Linearized free energy estimate

$$\frac{d\mathcal{E}}{dt} = -\frac{1}{\tau_0}\mathcal{I} + \mathcal{R}.$$

- **linearized** free energy

$$\mathcal{E} = \frac{1}{2}\int \frac{|f - f_\infty|^2}{f_\infty} dxdv + \frac{1}{2T_0}\|E\|_{L_x^2}^2.$$

- dissipation from **collision**

$$\mathcal{I} = T_0 \int \left|\partial_v\left(\frac{f}{f_\infty}\right)\right|^2 f_\infty dxdv.$$

- residual from **nonlinear field interaction**

$$\mathcal{R} = \frac{1}{2}\int E(f - f_\infty)\partial_v\left(\frac{f}{f_\infty}\right) dxdv.$$

Close-to-equilibrium framework: from linearized to nonlinear model

Linearized model

$$\partial_t f + v\partial_x f + E\partial_v f_\infty = \frac{1}{\tau_0}\partial_v\left(vf + T_0\partial_v f\right),$$

$$E = -\partial_x\Phi, \quad -\partial_{xx}\Phi = \rho - 1.$$

- free energy estimate

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = -\frac{1}{\tau_0}\mathcal{I}.$$

- $\mathcal{H}$ with $\lambda \sim \min(\tau_0, \tau_0^{-1})$

$$\mathcal{H} \sim \mathcal{E} \quad \textit{and} \quad \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} \leq -\lambda\mathcal{H}.$$

- exponential decay

$$\mathcal{E}(t) \lesssim e^{-\lambda t}\mathcal{E}(0).$$

Nonlinear model

$$\partial_t f + v\partial_x f + E\partial_v f_\infty + E\partial_v(f - f_\infty) = \frac{1}{\tau_0}\partial_v\left(vf + T_0\partial_v f\right),$$

$$E = -\partial_x\Phi, \quad -\partial_{xx}\Phi = \rho - 1.$$

- linearized free energy estimate

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = -\frac{1}{\tau_0}\mathcal{I} + \mathcal{R}.$$

- $\mathcal{H}$ with $\lambda \sim \min(\tau_0, \tau_0^{-1})$ and $\mu \sim \max(\tau_0, \tau_0^{-1})$

$$\mathcal{H} \sim \mathcal{E} \quad \textit{and} \quad \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} \leq -\lambda\mathcal{H} + \mu\mathcal{H}^2.$$

- exponential decay when $\mathcal{E}(0) \lesssim \min(\tau_0^2, \tau_0^{-2})$

$$\mathcal{E}(t) \lesssim e^{-\lambda t}\mathcal{E}(0).$$

Velocity discretization: scaled Hermite spectral method

Scaled Hermite spectral method

Scaled Hermite expansion

$$f(t,x,v) = \sum_{k\geq 0} D_k(t,x) H_k\left(\frac{v}{\sqrt{T_0}}\right) M(v).$$

Hermite system: $0 \leq k \leq N_H - 1$

$$\partial_t D_k + \sqrt{k}\mathcal{A} D_{k-1} - \sqrt{k+1}\mathcal{A}^* D_{k+1} - \sqrt{\frac{k}{T_0}} E D_{k-1} = -\frac{k}{T_0} D_k,$$

$$E = -\mathcal{B}\Phi, \qquad -\mathcal{B}^* E = D_0 - D_{\infty,0}.$$

where $\mathcal{A} = \sqrt{T_0}\partial_x$, $\mathcal{A}^* = -\sqrt{T_0}\partial_x$, $\mathcal{B} = \partial_x$ and $\mathcal{B}^* = -\partial_x$.

Translation

Linearized free energy

$$\mathcal{E} = \frac{1}{2}\sum_{k=0}^{N_H-1} \|D_k - D_{\infty,k}\|_{L_x^2}^2 + \frac{1}{2T_0}\|E\|_{L_x^2}^2.$$

Dissipation from **collision**:

$$\mathcal{I} = \sum_{k=1}^{N_H-1} k \left\|D_k\right\|_{L_x^2}^2$$

Residual from **field interaction**:

$$\mathcal{R} = \sum_{k=0}^{N_H-1} \sqrt{\frac{k+1}{T_0}} \langle E_h(D_{h,k} - D_{\infty,k}), D_{h,k+1}\rangle_{L_x^2}$$

Spatial discretization: structure-preserving conditions

Semidiscrete scheme

$$\partial_t D_{h,0} - \mathcal{A}_h^* D_{h,1} = 0,$$

$$\partial_t D_{h,1} + \mathcal{A}_h D_{h,0} - \sqrt{2}\mathcal{A}_h^* D_{h,2} + \mathcal{A}_h \Phi_h - \frac{1}{T_0} E_h (D_{h,0} - 1) = -\frac{1}{T_0} D_{h,1},$$

$$\partial_t D_{h,k} + \sqrt{k}\mathcal{A}_h D_{h,k-1} - \sqrt{k+1}\mathcal{A}_h^* D_{h,k+1} - \frac{k}{T_0} E_h D_{h,k-1} = -\frac{k}{T_0} D_{h,k}, k \geq 2$$

$$E_h = -\mathcal{B}_h \Phi_h, \quad -\mathcal{B}_h^* E_h = D_{h,0} - D_{\infty,0}.$$

Preservation of free energy estimate

Sufficient conditions

(H1) Duality: $\langle \mathcal{A}_h u, v \rangle_{L_x^2} = \langle u, \mathcal{A}_h^* v \rangle_{L_x^2}$.

(H2) Kernel: $\mathcal{A}_h 1 = 0$.

(H3) Duality: $\langle \mathcal{B}_h u, v \rangle_{L_x^2} = \langle u, \mathcal{B}_h^* v \rangle_{L_x^2}$.

$$\frac{d\mathcal{E}_h}{dt} = -\frac{1}{T_0}\mathcal{I}_h + \mathcal{R}_h.$$

Preservation of hypocoercivity

Sufficient conditions: (H1)-(H3) and

(H4) Poincaré inequality:

$$\|u\|_{L_x^2} \lesssim \|\mathcal{A}_h u\|_{L_x^2}.$$

(H5) Primal-dual balance:

$$\|\mathcal{A}_h u\|_{L_x^2} \sim \|\mathcal{A}_h^* u\|_{L_x^2}.$$

(H6) L^∞ regularity:

$$\|E_h\|_{L_x^\infty} \lesssim \|D_{h,0} - D_{\infty,0}\|_{L_x^2}.$$

If $\mathcal{E}_h(0) \leq \lambda^2$, then

$$\mathcal{E}_h(t) \leq 3e^{-\lambda t}\mathcal{E}_h(0), \quad \forall t \geq 0.$$

Spatial discretization: discontinuous Galerkin method

DG space

Quasi uniform mesh $\mathcal{T}_h$ for domain $\mathbb{T}$ with

$$h \le C_{qu} \min_{K \in \mathcal{T}_h} h_K.$$

Piecewise polynomial space

$$V_h^m = \left\{ u\, ;\, u\big|_K \in \mathscr{P}^m(K),\ \forall K \in \mathcal{T}_h \right\}.$$

Preservation of hypocoercivity

VFP equation: $(\mathcal{A}_h, \mathcal{A}_h^*)$

- **LDG** with alternating fluxes.

Poisson equation: $(\mathcal{B}_h, \mathcal{B}_h^*)$

- **LDG** with alternating fluxes.
- **Raviart-Thomas** mixed finite element.

Dependencies of decay rate

$$\lambda = \kappa \min(\tau_0, \tau_0^{-1}).$$

The constant κ depends ONLY on the **background temperature** T_0, the **domain length** $|\mathbb{T}|$, the **polynomial degree** m, and the **mesh quasi-uniformity** C_{qu}.

Temporal discretization: splitting method

Linearized step

$$\frac{D_{h,0}^{n+\frac{1}{2}} - D_{h,0}^{n}}{\Delta t} - \mathcal{A}_h^* D_{h,1}^{n+\frac{1}{2}} = 0,$$

$$\frac{D_{h,1}^{n+\frac{1}{2}} - D_{h,1}^{n}}{\Delta t} + \mathcal{A}_h D_{h,0}^{n+\frac{1}{2}} - \sqrt{2}\mathcal{A}_h^* D_{h,2}^{n+\frac{1}{2}} + \mathcal{A}_h \Phi_h^{n+\frac{1}{2}} = -\frac{1}{\tau_0} D_{h,1}^{n+\frac{1}{2}},$$

$$\frac{D_{h,k}^{n+\frac{1}{2}} - D_{h,k}^{n}}{\Delta t} + \sqrt{k}\mathcal{A}_h D_{h,k-1}^{n+\frac{1}{2}} - \sqrt{k+1}\mathcal{A}_h^* D_{h,k+1}^{n+\frac{1}{2}} = -\frac{k}{\tau_0} D_{h,k}^{n+\frac{1}{2}}, k \geq 2,$$

$$E_h^{n+\frac{1}{2}} = -\mathcal{B}_h \Phi_h^{n+\frac{1}{2}}, \quad -\mathcal{B}_h^* E_h^{n+\frac{1}{2}} = D_{h,0}^{n+\frac{1}{2}} - D_{\infty,0}.$$

Nonlinear step

$$D_{h,0}^{n+1} = D_{h,0}^{n+\frac{1}{2}},$$

$$\frac{D_{h,1}^{n+1} - D_{h,1}^{n+\frac{1}{2}}}{\Delta t} - \frac{1}{T_0} E_h^{n+1}(D_{h,0}^{n+1} - 1) = 0,$$

$$\frac{D_{h,k}^{n+1} - D_{h,k}^{n+\frac{1}{2}}}{\Delta t} - \frac{k}{T_0} E_h^{n+1} D_{h,k-1}^{n+1} = 0, k \geq 2,$$

$$E_h^{n+1} = E_h^{n+\frac{1}{2}}, \quad \Phi_h^{n+1} = \Phi_h^{n+\frac{1}{2}}.$$

Linear control

$$\frac{\mathcal{E}_h^{n+\frac{1}{2}} - \mathcal{E}_h^{n}}{\Delta t} \leq -\kappa \min(\tau_0, \tau_0^{-1}) \mathcal{E}_h^{n+\frac{1}{2}}.$$

Quadratic control

$$\frac{\mathcal{E}_h^{n+1} - \mathcal{E}_h^{n+\frac{1}{2}}}{\Delta t} \leq \kappa \max(\tau_0, \tau_0^{-1}) (\mathcal{E}_h^{n+1})^2.$$

Contents

Landau damping

Problem setup

Let $x \in \mathbb{T} = [0, L]$, $v \in \mathbb{R}$, and

$$f^{in}(x, v) = \left(1 + \delta \cos \frac{2\pi x}{L}\right) M(v),$$

with length of domain L, amplitude of perturbation δ, and background temperature $T_0 = 1$. In the collisionless limit $\tau_0 \to \infty$, this is the **classical Landau damping** configuration.

Solver setup

- second-order LDG with alternating fluxes for both VFP and Poisson in space;
- **Strang splitting** with second-order DIRK in time;
- **Hou-Li filter** with $2/3$ dealiasing;
- collisional regimes $\tau_0 \in \{10^1, 10^3, 10^5\}$.

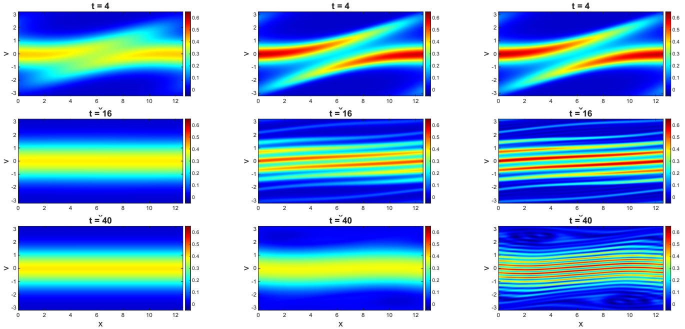
Landau damping: long-time behavior across collisional regimes

Strong collision

Collision dominates and the solution **relaxes rapidly toward equilibrium**.

Weak collision

Transport dominates for a long transient and generates **filamentation** in phase space.



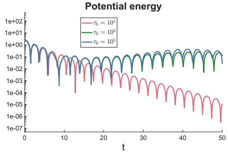
(a) $\tau_0 = 10^1$ (b) $\tau_0 = 10^3$ (c) $\tau_0 = 10^5$

Figure 1: Strong Landau damping ($\delta = 0.05$): snapshots of the distribution function f at $t \in \{4, 16, 40\}$ across the collisional regimes. $N_x = 128$, $N_H = 640$.

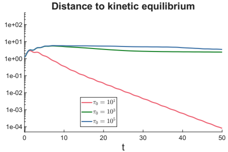
Landau damping: long-time convergence rate

Strong collision

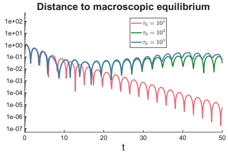
The numerical decay follows the **exponential** trend predicted by the discrete hypocoercive estimate with **close-to-equilibrium** initial data.



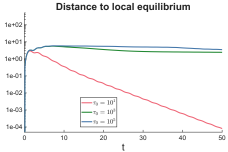
(a) $\|E\|_{L^2_x}$



(b) $\left\|f - f_\infty\right\|_{L^2_{x,v}(f_\infty^{-1})}$



(c) $\left\|\rho - \rho_\infty\right\|_{L^2_x(\rho_\infty^{-1})}$



(d) $\|f - \rho\mathcal{M}\|_{L^2_{x,v}(f_\infty^{-1})}$

Figure 2: Strong Landau damping ($\delta = 0.05$): time evolution in logarithmic scale of distances to equilibria across the collisional regimes. $N_x = 128$, $N_H = 640$.

Landau damping: accuracy test

Table 1: L^1, L^2 and L^∞ errors and convergence order, $\delta = 0.05$, $\Delta t = 0.1$, $t_{final} = 1$.

τ_0	$N_x \times N_H$	L^1 error	Order	L^2 error	Order	L^∞ error	Order
10^1	64 × 64	5.52E-03	-	1.82E-03	-	1.10E-03	-
	128 × 128	1.43E-03	1.95	4.70E-04	1.95	2.85E-04	1.96
	256 × 256	3.63E-04	1.98	1.20E-04	1.98	7.20E-05	1.98
10^3	64 × 64	5.87E-03	-	1.96E-03	-	1.23E-03	-
	128 × 128	1.31E-03	2.16	4.45E-04	2.14	2.89E-04	2.09
	256 × 256	2.67E-04	2.29	8.98E-05	2.31	5.62E-05	2.36
10^5	64 × 64	5.87E-03	-	1.96E-03	-	1.23E-03	-
	128 × 128	1.31E-03	2.16	4.44E-04	2.14	2.90E-04	2.09
	256 × 256	2.68E-04	2.29	8.98E-05	2.31	5.64E-05	2.36

Model with non-uniform ion density

Model reformulation

Model with ion density $\rho_0(x)$

$$\partial_t f + v\partial_x f + E\partial_v f = \frac{1}{T_0}\partial_v(vf + T_0\partial_v f),$$

$$E = -\partial_x\Phi, -\partial_{xx}\Phi = \rho - \rho_0(x),$$

Let $F = E - E_\infty, \Psi = \Phi - \Phi_\infty$, then

$$\partial_t f + v\partial_x f + E_\infty\partial_v f + F\partial_v f = \frac{1}{T_0}\partial_v(vf + T_0\partial_v f),$$

$$F = -\partial_x\Psi, -\partial_{xx}\Psi = \rho - \rho_\infty,$$

Gibbs state

Poisson-Boltzmann equation:

$$\rho_\infty = c_\infty \exp(-\frac{\Phi_\infty}{T_0}), \quad -\Delta_x\Phi_\infty = \rho_\infty - \rho_0,$$

where c_∞ is determined by

$$c_\infty \int \exp(-\frac{\Phi_\infty}{T_0})dx = \int \rho_0(x)dx.$$

Landau damping with non-uniform ion density

Problem setup

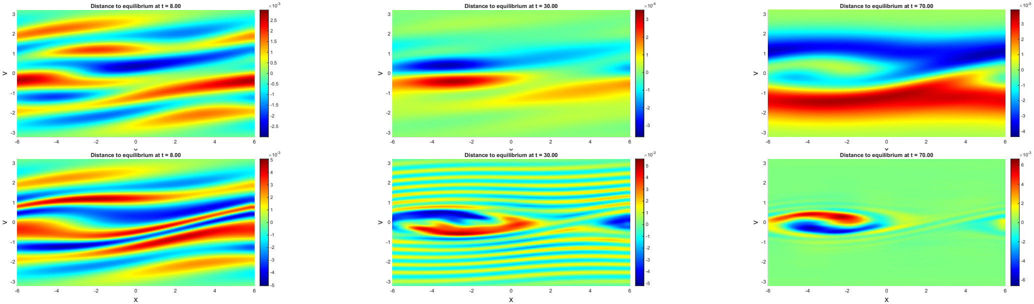
Let $x \in \mathbb{T} = [-L, L]$ with $L = 6$, $v \in \mathbb{R}$, and consider initial data

$$f^{in}(x, v) = (\rho_\infty(x) + \delta cos(\frac{\pi}{L}x))M(v)),$$

where amplitude of perturbation $\delta = 0.01$ and background temperature $T_0 = 1$. Consider stationary density ρ_∞ be given by

$$\rho_\infty(x) = 0.2\,\sin(\frac{\pi}{L}x).$$

Landau damping with non-uniform ion density



(a) $t = 8$ (b) $t = 30$ (c) $t = 70$

Figure 3: Landau damping with non-uniform ion density: snapshots of **distance** between distribution f and the equilibrium f_∞ at $t \in \{8, 30, 70\}$ with $(N_x, N_H) = (128, 400)$. The first line is for strong collision $\tau_0 = 10^2$ (max distance is about 10^{-5} at $t = 70$), while the second line is for weak collision $\tau_0 = 10^4$ (max distance is about 10^{-3} at $t = 70$).

Two-stream instability with non-uniform ion density

Problem setup

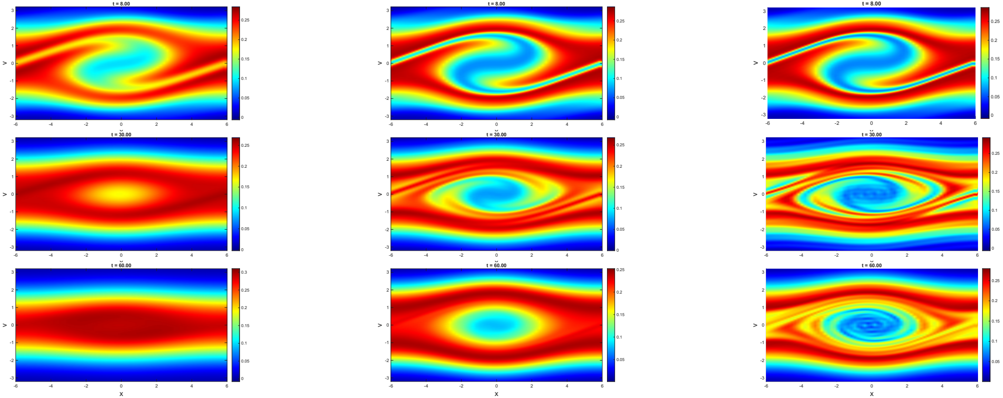
Let $x \in \mathbb{T} = [-L, L]$ with $L = 6$, $v \in \mathbb{R}$, and consider initial data

$$f^{in}(x,v) = (\frac{1}{6} + \frac{5}{6}|v|^2)(1 + \delta cos(\frac{\pi}{L}x))M(v)),$$

where amplitude of perturbation $\delta = 0.01$ and background temperature $T_0 = 1$. Consider stationary potential Φ_∞ given by

$$\Phi_\infty(x) = 0.1(1 - \cos(\frac{\pi}{L}x)).$$

Two-stream instability with non-uniform ion density



(a) $\tau_0 = 10^2$ (b) $\tau_0 = 10^3$ (c) $\tau_0 = 10^4$

Figure 4: Two stream instability with non-uniform density: snapshots of distribution function f at $t \in \{8, 30, 60\}$ (from top to bottom) across the collisional regimes. $N_x = 128$, $N_H = 800$.

Contents

Conclusions

Hypocoercive mechanism

microscopic dissipation + transport coupling $\Longrightarrow$ macroscopic dissipation.

VPFP solver

- **sufficient conditions** for preserving hypocoercivity at the **semidiscrete** level.
- **provable** hypocoercivity-preserving DG methods: exponential relaxation for close-to-equilibrium initial data.

Perspectives

Analysis

- extension to high-dimensional case.
- extension to other kinetic models, e.g., BGK equation and radiative transfer equation.

Algorithm

- efficient high-order hypocoercivity-preserving time integrators.
- diffusive regime.
- dynamic scaling in Hermite expansion.

References

- A. Blaustein and F. Filbet, On a discrete framework of hypocoercivity for kinetic equations, **Mathematics of Computation** 93 no. 345 (2024) 163–202.
- A. Blaustein and F. Filbet, A structure and asymptotic preserving scheme for the Vlasov-Poisson-Fokker-Planck model, **Journal of Computational Physics** 498 (2024) 112693.
- **Y. Cai**, A. Blaustein, T. Xiong and F. Filbet, Discrete hypocoercive estimates for discontinuous Galerkin methods: application to the Vlasov-Poisson-Fokker-Planck system, **arXiv:2603.26218** (2026).

Thank you for your attention!